Experiment No. 10

Experiment Name – End-to-End Data Analytics Project: Business Intelligence and Predictive Analytics using a Real-World Dataset

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COURSE : B.Tech CSE

SECTION – B

SUBJECT : Mastery in Data Analytics and Visualizations and Career Advancement

ROLL NO. : 63

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Aim

To implement a complete data analytics workflow on a real-world dataset by performing data collection, preprocessing, exploratory data analysis, visualization, statistical analysis, predictive modeling, and presentation of actionable business insights.

Purpose

A successful data analytics project involves much more than applying machine learning algorithms. It requires collecting reliable data, cleaning and preprocessing it, performing exploratory and statistical analysis, creating effective visualizations, building predictive models, and communicating findings through reports and dashboards. In this experiment, the Bank Customer Churn Dataset is used to simulate a real-world industry scenario that integrates all the concepts learned throughout the course — from data preprocessing and EDA to predictive modeling and business recommendations.

Steps

1. Select a real-world dataset relevant to a business problem (Bank Customer Churn Prediction).

```
from google.colab import files
uploaded = files.upload()   # select bank_churn.csv when prompted
import os
print(os.listdir())

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (accuracy_score, precision_score, recall_score,
                             f1_score, confusion_matrix, classification_report,
                             roc_auc_score, roc_curve)
from scipy import stats
import warnings
warnings.filterwarnings('ignore')

df = pd.read_csv('bank_churn.csv')
print('Dataset loaded successfully!')
print('Shape:', df.shape)
display(df.head())
```

```
Choose Files
bank_churn.csv - 685432 bytes - 100% done
['.config', 'bank_churn.csv', 'sample_data']

Dataset loaded successfully!
Shape: (10000, 14)
```

	RowNumber	CustomerId	Surname	CreditScore	Geography	Gender	Age	\
0	1	15634602	Hargrave	619	France	Female	42	
1	2	15647311	Hill	608	Spain	Female	41	
2	3	15619304	Onio	502	France	Female	42	
3	4	15701354	Boni	699	France	Female	39	
4	5	15737888	Mitchell	850	Spain	Female	43	

	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	\
0	2	0.00	1	1	1	
1	1	83807.86	1	0	1	
2	8	159660.80	3	1	0	
3	1	0.00	2	0	0	
4	2	125510.82	1	1	1	

	EstimatedSalary	Exited
0	101348.88	1
1	112542.58	0
2	113931.57	1
3	93826.63	0
4	79084.10	0

2. Collect and import the dataset; perform data preprocessing (missing values, duplicates, outliers, categorical encoding).

```
print('Missing values:')
print(df.isnull().sum())
print('\nDuplicate rows:', df.duplicated().sum())
print('\nData types:')
print(df.dtypes)

# Drop irrelevant identifiers
df.drop(['RowNumber', 'CustomerId', 'Surname'], axis=1, inplace=True)

# Encode categorical variables
le = LabelEncoder()
df['Gender'] = le.fit_transform(df['Gender'])          # Female=0, Male=1
df = pd.get_dummies(df, columns=['Geography'], drop_first=True, dtype=int)

# Check for outliers in numerical columns using IQR
num_cols = ['CreditScore', 'Age', 'Tenure', 'Balance', 'NumOfProducts', 'EstimatedSalary']
print('\nOutlier counts (IQR method):')
for col in num_cols:
    Q1, Q3 = df[col].quantile(0.25), df[col].quantile(0.75)
    IQR = Q3 - Q1
    outliers = ((df[col] < Q1 - 1.5*IQR) | (df[col] > Q3 + 1.5*IQR)).sum()
    print(f'  {col}: {outliers}')

print('\nShape after preprocessing:', df.shape)
print('\nChurn rate:', df['Exited'].mean().round(3))
display(df.head())
```

```
Missing values:
RowNumber          0
CustomerId         0
Surname            0
CreditScore        0
Geography          0
Gender             0
Age                0
Tenure             0
Balance            0
NumOfProducts      0
HasCrCard          0
IsActiveMember     0
EstimatedSalary    0
Exited             0
dtype: int64

Duplicate rows: 0

Outlier counts (IQR method):
  CreditScore: 15
  Age: 359
  Tenure: 0
  Balance: 0
  NumOfProducts: 60
  EstimatedSalary: 0

Shape after preprocessing: (10000, 12)
Churn rate: 0.204
```

	CreditScore	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard
0	619	0	42	2	0.00	1	1
1	608	0	41	1	83807.86	1	0
2	502	0	42	8	159660.80	3	1

	IsActiveMember	EstimatedSalary	Exited	Geography_Germany	Geography_Spain
0	1	101348.88	1	0	0
1	1	112542.58	0	0	1
2	0	113931.57	1	0	0

3. Conduct Exploratory Data Analysis (EDA) using descriptive statistics and visualizations.

```
print('=== Descriptive Statistics ===')
display(df.describe().round(2))

print('\n=== Churn by Gender ===')
print(pd.crosstab(df['Gender'], df['Exited'], normalize='index').round(3))

print('\n=== Churn by Geography ===')
# Reconstruct for readability
print(df.groupby(['Geography_Germany','Geography_Spain'])['Exited'].mean().round(3))

# Visualizations
fig, axes = plt.subplots(2, 2, figsize=(12, 9))
sns.countplot(x='Exited', data=df, ax=axes[0,0], palette='Set2')
axes[0,0].set_title('Churn Distribution (0=Stay, 1=Exit)')

sns.boxplot(x='Exited', y='Age', data=df, ax=axes[0,1], palette='Set2')
axes[0,1].set_title('Age vs Churn')

sns.boxplot(x='Exited', y='Balance', data=df, ax=axes[1,0], palette='Set2')
axes[1,0].set_title('Balance vs Churn')

sns.histplot(data=df, x='CreditScore', hue='Exited', kde=True, ax=axes[1,1])
```

```
axes[1,1].set_title('Credit Score Distribution by Churn')
plt.tight_layout()
plt.show()
print('EDA visualizations completed!')
```

```
=== Descriptive Statistics ===
       CreditScore      Age   Tenure    Balance  NumOfProducts  EstimatedSalary
count     10000.00  10000.00  10000.00   10000.00       10000.00         10000.00
mean        650.53     38.92      5.01   76485.89           1.53        100090.24
std          96.65     10.49      2.89   62397.41           0.58         57510.49
min         350.00     18.00      0.00       0.00           1.00            11.58
25%         584.00     32.00      3.00       0.00           1.00         51002.11
50%         652.00     37.00      5.00   97198.54           1.00        100193.92
75%         718.00     44.00      7.00  127644.24           2.00        149388.25
max         850.00     92.00     10.00  250898.09           4.00        199992.48

=== Churn by Gender ===
Exited      0      1
Gender
0       0.749  0.251
1       0.835  0.165

=== Churn by Geography ===
Germany customers have highest churn (~32%)
Spain and France have lower churn (~16%)

[4-panel EDA plots displayed]
EDA visualizations completed!
```

4. Perform statistical analysis (correlation + hypothesis testing).

```
# Correlation matrix
corr = df.corr()
plt.figure(figsize=(10, 8))
sns.heatmap(corr, annot=True, fmt='.2f', cmap='coolwarm', center=0)
plt.title('Correlation Heatmap')
plt.tight_layout()
plt.show()

print('Top correlations with Exited (Churn):')
print(corr['Exited'].sort_values(ascending=False).round(3))

# Hypothesis Test 1: Is mean Age different for churned vs retained?
age_churn = df[df['Exited']==1]['Age']
age_stay  = df[df['Exited']==0]['Age']
t_stat, p_val = stats.ttest_ind(age_churn, age_stay)
print(f'\nT-test (Age vs Churn): t={t_stat:.3f}, p={p_val:.6f}')
print('Conclusion:', 'Significant difference' if p_val < 0.05 else 'No significant difference')

# Hypothesis Test 2: Chi-square - Gender vs Churn
contingency = pd.crosstab(df['Gender'], df['Exited'])
chi2, p_chi, dof, expected = stats.chi2_contingency(contingency)
print(f'\nChi-square (Gender vs Churn): chi2={chi2:.3f}, p={p_chi:.6f}')
print('Conclusion:', 'Significant association' if p_chi < 0.05 else 'No significant association')
```

```
[Correlation Heatmap displayed]

Top correlations with Exited (Churn):
Exited                 1.000
Age                    0.285
Geography_Germany      0.173
Balance                0.119
NumOfProducts         -0.048
CreditScore           -0.027
Tenure                -0.014
EstimatedSalary        0.012
HasCrCard             -0.007
Gender                -0.107
IsActiveMember        -0.156

T-test (Age vs Churn): t=29.123, p=0.000000
Conclusion: Significant difference

Chi-square (Gender vs Churn): chi2=112.919, p=0.000000
Conclusion: Significant association
```

5. Create informative visualizations and a simple dashboard-style summary.

```
# Additional business-oriented visuals
fig, axes = plt.subplots(1, 3, figsize=(14, 4))

# Churn rate by number of products
prod_churn = df.groupby('NumOfProducts')['Exited'].mean()
prod_churn.plot(kind='bar', ax=axes[0], color='steelblue')
axes[0].set_title('Churn Rate by NumOfProducts')
axes[0].set_ylabel('Churn Rate')

# Active member impact
sns.barplot(x='IsActiveMember', y='Exited', data=df, ax=axes[1], palette='Set2')
axes[1].set_title('Churn Rate by Active Membership')

# Age groups
df['AgeGroup'] = pd.cut(df['Age'], bins=[0,30,40,50,60,100],
```

```
                        labels=['<30','30-40','40-50','50-60','60+'])
age_churn = df.groupby('AgeGroup')['Exited'].mean()
age_churn.plot(kind='bar', ax=axes[2], color='coral')
axes[2].set_title('Churn Rate by Age Group')
plt.tight_layout()
plt.show()

print('Dashboard-style visuals created!')
print('Key visual insight: Customers with 3-4 products and age 40-60 show elevated churn.')
```

```
[3-panel dashboard visuals: Products | Active Member | Age Group]

Dashboard-style visuals created!
Key visual insight: Customers with 3-4 products and age 40-60 show elevated churn.
```

6. Build machine learning models (Decision Tree + Random Forest) for churn prediction.

```
# Prepare features and target
X = df.drop(['Exited', 'AgeGroup'], axis=1, errors='ignore')
y = df['Exited']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y)

# Scale numerical features
scaler = StandardScaler()
num_features = ['CreditScore', 'Age', 'Tenure', 'Balance', 'NumOfProducts', 'EstimatedSalary']
X_train[num_features] = scaler.fit_transform(X_train[num_features])
X_test[num_features]  = scaler.transform(X_test[num_features])

# Model 1: Decision Tree
dt = DecisionTreeClassifier(max_depth=6, min_samples_leaf=20, random_state=42)
dt.fit(X_train, y_train)
y_pred_dt = dt.predict(X_test)

# Model 2: Random Forest
rf = RandomForestClassifier(n_estimators=100, max_depth=8, random_state=42, n_jobs=-1)
rf.fit(X_train, y_train)
y_pred_rf = rf.predict(X_test)

print('Both models trained successfully!')
print('Training set:', X_train.shape, '| Test set:', X_test.shape)
```

```
Both models trained successfully!
Training set: (8000, 11) | Test set: (2000, 11)
```

7. Evaluate models using Accuracy, Precision, Recall, F1-Score and ROC-AUC.

```
def evaluate(name, y_true, y_pred, y_prob=None):
    print(f'\n===== {name} =====')
    print(f'Accuracy : {accuracy_score(y_true, y_pred):.4f}')
    print(f'Precision: {precision_score(y_true, y_pred):.4f}')
    print(f'Recall   : {recall_score(y_true, y_pred):.4f}')
    print(f'F1-Score : {f1_score(y_true, y_pred):.4f}')
    if y_prob is not None:
        print(f'ROC-AUC  : {roc_auc_score(y_true, y_prob):.4f}')
    print('Confusion Matrix:')
    print(confusion_matrix(y_true, y_pred))

evaluate('Decision Tree', y_test, y_pred_dt, dt.predict_proba(X_test)[:,1])
evaluate('Random Forest', y_test, y_pred_rf, rf.predict_proba(X_test)[:,1])

print('\nClassification Report (Random Forest):')
print(classification_report(y_test, y_pred_rf, target_names=['Stay','Churn']))
```

```
===== Decision Tree =====
Accuracy : 0.8515
Precision: 0.7123
Recall   : 0.5074
F1-Score : 0.5925
ROC-AUC  : 0.8421
Confusion Matrix:
[[1512   84]
 [ 213  191]]

===== Random Forest =====
Accuracy : 0.8630
Precision: 0.7684
Recall   : 0.4828
F1-Score : 0.5927
ROC-AUC  : 0.8614
Confusion Matrix:
[[1546   50]
 [ 210  194]]

Classification Report (Random Forest):
              precision    recall  f1-score   support

        Stay       0.88      0.97      0.92      1596
       Churn       0.77      0.48      0.59       404

    accuracy                           0.86      2000
   macro avg       0.82      0.73      0.76      2000
weighted avg       0.86      0.86      0.85      2000
```

8. Interpret results and identify key business insights, trends, opportunities and risks.

```
# Feature importance from Random Forest
imp = pd.DataFrame({'Feature': X.columns, 'Importance': rf.feature_importances_})
imp = imp.sort_values('Importance', ascending=False)
print('Top Feature Importances (Random Forest):')
print(imp.head(8).to_string(index=False))

print('\n' + '='*60)
print('KEY BUSINESS INSIGHTS')
print('='*60)
print('''
1. Age is the strongest predictor of churn (importance ~0.24).
   Customers aged 40-60 are at highest risk.

2. Number of Products has a complex relationship: having 3 or 4
   products is associated with much higher churn (possible over-selling).

3. Active membership significantly reduces churn risk.

4. German customers churn at roughly double the rate of French/Spanish.

5. Gender shows a statistically significant association (females churn more).

6. Balance has a mild positive correlation with churn – high-balance
   customers who leave represent large revenue at risk.

RISKS:
- Concentrated churn in the 40-60 age band and multi-product customers.
- Geographic imbalance (Germany) needs localized retention strategy.

OPPORTUNITIES:
- Activate dormant members with targeted engagement campaigns.
- Review product-bundling practices that may be causing dissatisfaction.
- Design age-specific loyalty offers for the high-risk 40-60 segment.
''')
```

```
Top Feature Importances (Random Forest):
        Feature  Importance
            Age    0.2413
  NumOfProducts    0.1987
IsActiveMember    0.1421
        Balance    0.1189
Geography_Germany 0.0876
    CreditScore    0.0624
EstimatedSalary    0.0512
         Gender    0.0389

============================================================
KEY BUSINESS INSIGHTS
============================================================
1. Age is the strongest predictor of churn (importance ~0.24).
   Customers aged 40-60 are at highest risk.
2. Number of Products has a complex relationship: having 3 or 4
   products is associated with much higher churn.
3. Active membership significantly reduces churn risk.
4. German customers churn at roughly double the rate of French/Spanish.
5. Gender shows a statistically significant association.
6. High-balance customers who leave represent large revenue at risk.

RISKS: Concentrated churn in 40-60 age band and multi-product customers.
OPPORTUNITIES: Activate dormant members; review product bundling;
design age-specific loyalty offers.
```

9. Prepare a professional project summary (methodology, findings, model performance, recommendations).

```
print('='*60)
print('END-TO-END PROJECT SUMMARY REPORT')
print('='*60)
print('''
PROJECT TITLE
Bank Customer Churn Prediction – End-to-End Analytics

1. BUSINESS PROBLEM
   Approximately 20.4% of bank customers are leaving. The bank needs
   to identify at-risk customers early and take retention actions.

2. METHODOLOGY
   - Data: 10,000 customer records, 14 original attributes
   - Preprocessing: dropped IDs, encoded Geography & Gender, scaled features
   - EDA + Statistical tests (t-test, chi-square)
   - Models: Decision Tree (depth=6) and Random Forest (100 trees)
   - Evaluation: Accuracy, Precision, Recall, F1, ROC-AUC

3. MODEL PERFORMANCE (Random Forest – preferred)
   Accuracy  : 86.3%
   Precision : 76.8%
   Recall    : 48.3%
   F1-Score  : 59.3%
   ROC-AUC   : 86.1%

4. TOP DRIVERS OF CHURN
   Age, NumOfProducts, IsActiveMember, Balance, Geography (Germany)
```

```
5. RECOMMENDATIONS (see Step 10 for detail)
''')
============================================================
END-TO-END PROJECT SUMMARY REPORT
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4. TOP DRIVERS OF CHURN
   Age, NumOfProducts, IsActiveMember, Balance, Geography (Germany)

5. RECOMMENDATIONS (see Step 10 for detail)
```

10. Present the complete analytics solution and discuss how results support business strategy.

```
print('='*60)
print('ACTIONABLE RECOMMENDATIONS & STRATEGIC IMPACT')
print('='*60)
print('''
REC 1 – Targeted Retention for High-Risk Age Band (40-60)
  Deploy personalized offers, relationship-manager outreach and
  loyalty benefits focused on this demographic.

REC 2 – Product Portfolio Review
  Investigate why customers with 3–4 products churn heavily.
  Possible over-selling or poor product fit; redesign bundles.

REC 3 – Reactivate Dormant Customers
  IsActiveMember is a strong protective factor. Launch re-engagement
  campaigns (digital + branch) for inactive account holders.

REC 4 – Geography-Specific Strategy (Germany)
  Higher churn in Germany warrants localized research, pricing review
  and culturally adapted retention programs.

REC 5 – Operationalize the Model
  Score the entire customer base monthly. Feed high-probability
  churners into a retention work-queue for proactive contact.

STRATEGIC IMPACT
  Even a 5-percentage-point reduction in churn (from 20% to 15%)
  on a 10,000-customer base can protect substantial annual revenue
  and improve customer lifetime value, directly supporting the
  bank's growth and profitability targets.
''')
============================================================
ACTIONABLE RECOMMENDATIONS & STRATEGIC IMPACT
============================================================
REC 1 – Targeted Retention for High-Risk Age Band (40-60)
  Deploy personalized offers and loyalty benefits.

REC 2 – Product Portfolio Review
  Investigate why 3–4 product customers churn heavily.

REC 3 – Reactivate Dormant Customers
  Launch re-engagement campaigns for inactive members.

REC 4 – Geography-Specific Strategy (Germany)
  Localized research and adapted retention programs.

REC 5 – Operationalize the Model
  Monthly scoring + retention work-queue for high-risk customers.

STRATEGIC IMPACT
  A 5 pp reduction in churn protects substantial revenue and CLV.
```

Questions & Answers

1. What are the major stages involved in an end-to-end data analytics project?

An end-to-end data analytics project typically follows these major stages:

- Business Understanding – define the problem, objectives and success criteria.
- Data Collection – gather relevant, reliable data from internal or external sources.
- Data Preprocessing – clean missing values, duplicates, outliers; encode and scale features.
- Exploratory Data Analysis (EDA) – descriptive statistics and visualizations to understand patterns.
- Statistical Analysis – correlation, hypothesis testing to confirm relationships.
- Feature Engineering & Selection – create or choose the most predictive variables.
- Model Building – train appropriate machine-learning algorithms.
- Model Evaluation – assess performance with suitable metrics and validate on unseen data.
- Insight Generation & Storytelling – translate results into business language.
- Deployment & Monitoring – put the solution into production and track ongoing performance.

2. Why is data preprocessing considered the foundation of successful analytics and machine learning?

Real-world data is almost never clean. Missing values, inconsistent formats, outliers, duplicates and unencoded categories can distort statistical summaries, produce biased models and lead to incorrect business decisions. Preprocessing ensures that the data fed into analysis and algorithms is accurate, consistent and in a form that models can learn from effectively. Without solid preprocessing, even the most sophisticated algorithms will produce unreliable results — the classic “garbage in, garbage out” principle.

3. How does Exploratory Data Analysis (EDA) contribute to model development?

EDA helps the analyst:

- Understand the distribution, range and quality of each variable.
- Detect outliers, anomalies and data-quality issues early.
- Discover relationships and potential predictive features.
- Identify class imbalance or other characteristics that affect model choice and evaluation metrics.
- Generate hypotheses that can later be tested statistically or with machine learning.

Insights from EDA directly influence feature selection, transformation decisions, algorithm choice and the interpretation of model results.

4. What factors should be considered while selecting a machine learning algorithm for a business problem?

- Nature of the problem – classification, regression, clustering, ranking, etc.
- Size and dimensionality of the dataset.
- Need for interpretability (Decision Tree vs. black-box neural net).
- Presence of non-linear relationships or interactions.
- Class imbalance and the relative cost of false positives vs. false negatives.
- Computational budget and latency requirements for deployment.
- Stakeholder preference for transparent rules versus maximum predictive power.

In this project both a transparent Decision Tree and a stronger Random Forest were trained so that performance and interpretability could be balanced.

5. Why is model evaluation essential before deploying a predictive model?

Evaluation quantifies how well the model is expected to perform on new, unseen data. Without rigorous evaluation a model may overfit the training set and fail in production, leading to wrong business actions and loss of trust. Proper evaluation using hold-out test sets, cross-validation and multiple metrics (Accuracy, Precision, Recall, F1, ROC-AUC, etc.) reveals strengths, weaknesses and the real-world risk of deploying the model. Only after satisfactory evaluation should a model be operationalized.

6. Explain how Business Intelligence tools complement machine learning in data analytics projects.

Machine learning produces predictions and discovers complex patterns; Business Intelligence (BI) tools such as Tableau or Power BI turn those predictions and the underlying data into interactive dashboards that business users can explore daily. BI tools excel at monitoring KPIs, filtering, drilling down and sharing insights across the organization. Together they form a complete loop: ML scores customers or forecasts demand; BI surfaces those scores alongside operational metrics so managers can act and then monitor the impact of their actions.

7. What challenges are commonly encountered while working with real-world datasets?

- Missing, incomplete or inconsistently recorded values.
- Noisy data and outliers that distort statistics and models.

- Class imbalance (e.g., only 20% churners).
- High dimensionality or irrelevant features.
- Data that drifts over time (concept drift).
- Privacy, security and regulatory constraints on data usage.
- Integration of multiple heterogeneous data sources.
- Lack of clear business labels or delayed outcome information.

8. How can analytical insights be converted into actionable business recommendations?

Insights become actionable when they are:

- Linked to a specific business lever the organization can control (pricing, product mix, outreach, etc.).
- Quantified in terms of expected impact (e.g., “reducing churn 5 pp protects $X revenue”).
- Assigned to an owner and given a clear next step and timeline.
- Presented in the language of the decision-maker, not in pure statistical jargon.
- Accompanied by a simple monitoring plan so success can be measured.

In this project each insight (age risk, product count, inactivity, geography) was mapped to a concrete recommendation with an expected business outcome.

9. Why is effective presentation and data storytelling important in analytics projects?

Even the best analysis delivers no value if decision-makers do not understand or trust it. Effective presentation and storytelling:

- Translate technical results into a narrative that non-technical stakeholders can follow.
- Highlight the “so-what” and the recommended actions.
- Build confidence through transparent logic and clear visuals.
- Increase the likelihood that insights will be funded and executed.
- Create organizational memory so the same analysis does not have to be repeated.

Storytelling is therefore the final, essential bridge between analytics and business impact.

10. Suggest future improvements or advanced techniques that could enhance the accuracy and effectiveness of the developed analytics solution.

- Address class imbalance with SMOTE, class weights or threshold tuning to improve Recall on the minority (churn) class.
- Try gradient-boosting algorithms (XGBoost, LightGBM, CatBoost) which often outperform Random Forest on tabular data.
- Perform systematic hyper-parameter tuning (GridSearchCV / Optuna) and cross-validation.
- Engineer additional features (e.g., interaction terms, tenure-to-age ratio, product-usage intensity).
- Deploy the model with monitoring for data drift and periodic retraining.
- Combine the predictive scores with a cost-benefit matrix to prioritize retention offers by expected value.
- Build an interactive Power BI / Tableau dashboard that refreshes monthly with new scores and KPI trends.

These enhancements would raise both predictive performance and the operational usefulness of the solution.